

**This assignment will not be graded and is only for practice.**

### Key Points: 1

Let  $T : V \rightarrow W$  be a linear transformation. We define kernel of  $T$  (denoted by  $\ker(T)$ ) and image of  $T$  (denoted by  $\text{Im}(T)$ ) as follows:

$$\ker(T) = \{v \in V \mid T(v) = 0\}$$

$$\text{Im}(T) = \{w \in W \mid \text{there exists } v \in V \text{ for which } T(v) = w\}$$

$T$  is injective (i.e.,  $T$  is a monomorphism) if and only if  $\ker(T) = \{0\}$ .

A linear transformation  $T : V \rightarrow W$  is surjective (i.e.,  $T$  is epimorphism) if and only if  $\text{Im}(T) = W$ .

$\text{Nullity}(T) = \dim(\ker(T))$  and  $\text{Rank}(T) = \dim(\text{Im}(T))$ .

Let  $T : V \rightarrow W$  be a linear transformation. Then

$$\text{Rank}(T) + \text{Nullity}(T) = \dim(V).$$

This is known as the rank nullity theorem.

Let  $T : V \rightarrow W$  be a linear transformation.  $T$  is injective if and only if  $\text{Nullity}(T) = 0$ .  $T$  is surjective if and only if  $\text{Rank}(T) = \dim(W)$ .

Consider the following linear transformation:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer the following questions:

1) Which of the following Options is true?

**1 point**

[Hint:  $T(x, y, z) = (2x + 3z, 4y + z) = (0, 0)$ , i.e.,  $2x + 3z = 0$  and  $4y + z = 0$ ]

☐  $\ker(T) = \{(3y, y, -4y) \mid y \in \mathbb{R}\}$

☐  $\ker(T) = \{(6y, y, 4y) \mid y \in \mathbb{R}\}$

☐  $\ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$

☐  $\ker(T) = \{(3y, y, 4y) \mid y \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$

2) Which of the following is a basis of  $\text{Ker}(T)$ ?

**1 point**

- ☐  $\{(3, 1, -4)\}$
- ☐  $\{(6, 0, -4), (0, 1, -4)\}$
- ☐  $\{(6, 1, -4)\}$
- ☐  $\{(3, 0, -2), (0, 1, -4)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(6, 1, -4)\}$

3) Find the nullity of  $T$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

4) Find the rank of  $T$ .

[Hint: use rank nullity theorem.]

No, the answer is incorrect.

Score: 0

Feedback:

– Here  $\dim(V) = 3$  as  $V = \mathbb{R}^3$ .

Accepted Answers:

(Type: Numeric) 2

**1 point**

5) Which of the following is true?

**1 point**

- ☐  $T$  is both injective and surjective.
- ☐  $T$  is injective but not surjective.
- ☐  $T$  is surjective but not injective.
- ☐  $T$  is neither injective nor surjective.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is surjective but not injective.

6) Which option represents the kernel and image of the following linear transformation?

**1 point**

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x, 0)$$

- ☐  $\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$
- ☐  $\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}.$
- ☐  $\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$
- ☐  $\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}.$

No, the answer is incorrect.

Score: 0

Feedback:

•  $\text{Ker}(T) = \{(0, y) \mid y \in \mathbb{R}\}.$

Accepted Answers:

$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$

**Key Points: 2**

Let  $T : V \rightarrow W$  be a linear. Follow the steps below to find bases for the null space and the range space of  $T$ .

Step 1: Find the matrix  $A$  corresponding to  $T$  with respect to some standard ordered bases  $\beta = \{v_1, v_2, \dots, v_n\}$  and  $\gamma = \{w_1, w_2, \dots, w_m\}$  for  $V$  and  $W$  respectively.

Step 2: Use row reduction on  $A$  to obtain the matrix  $R$  which is in reduced row echelon form.

Step 3: The basis of the solution space of  $Rx = 0$  is the basis of null space of matrix  $A$  and can be obtained by finding the pivot and non-pivot columns (dependent and independent variables) as seen earlier.

Step 4: The vectors  $\begin{bmatrix} c_{11} \\ c_{12} \\ \vdots \\ c_{1n} \end{bmatrix}, \begin{bmatrix} c_{21} \\ c_{22} \\ \vdots \\ c_{2n} \end{bmatrix}, \dots, \begin{bmatrix} c_{k1} \\ c_{k2} \\ \vdots \\ c_{kn} \end{bmatrix}$  form a basis of the null space of  $A$  precisely when the vectors

$v'_1, v'_2, \dots, v'_k \in \ker(T)$  where  $v'_i = \sum_{j=1}^n c_{ij} v_j$ , form a basis for  $\ker(T)$ . Use the basis obtained in step 3 to thus get a basis for  $\ker(T)$ .

Step 5: Recall that if  $i_1, i_2, \dots, i_r$  are the columns of  $R$  containing the pivot elements, then the same columns of  $A$  form a basis for the column space of  $A$ .

Step 6: The vectors  $\begin{bmatrix} d_{11} \\ d_{12} \\ \vdots \\ d_{1m} \end{bmatrix}, \begin{bmatrix} d_{21} \\ d_{22} \\ \vdots \\ d_{2m} \end{bmatrix}, \dots, \begin{bmatrix} d_{r1} \\ d_{r2} \\ \vdots \\ d_{rm} \end{bmatrix}$  form a basis of the column space of  $A$  precisely when the

vectors  $w'_1, w'_2, \dots, w'_r \in \text{im}(T)$  where  $w'_i = \sum_{j=1}^m d_{ij} w_j$ , form a basis for  $\text{im}(T)$ . Use the basis obtained in step 5 to thus get a basis for  $\text{im}(T)$ .

Consider the following linear transformation:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z)$$

Answer the following questions 7-9 using the information given above.

7) Which of the matrices corresponds the given linear transformation  $T$  with respect to the standard ordered basis of  $\mathbb{R}^3$  for both the domain and the codomain?

**1 point**



$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$$

☐  $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 3 \\ 0 & 1 & 0 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

8) What will be the kernel of  $T$ ?

**1 point**

☐  $\{(x, 0, z), (0, y, z) \mid x, y, z \in \mathbb{R}\}.$

☐  $\{(z, 0, z), (0, -z, z) \mid z \in \mathbb{R}\}.$

☐  $\{(z, -z, z) \mid z \in \mathbb{R}\}.$

☐  $\{(x, 0, z), (0, -x, z) \mid x, z \in \mathbb{R}\}.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(z, -z, z) \mid z \in \mathbb{R}\}.$$

9) What will be the basis of the image space of  $T$  found by the algorithm mentioned in the Key Points ? **1 point**

- ☐  $\{(1, 0, -1), (2, 3, 1)\}$
- ☐  $\{(1, 0, 0), (0, 1, 0)\}$
- ☐  $\{(1, 2, 0), (0, 3, 3)\}$
- ☐  $\{(-1, 1, 3)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 2, 0), (0, 3, 3)\}$

Your score is: 0/9